3-STACK NANOSHEET GAAFET

TEAM EXPERIMENT WORKPLAN

Physics-informed AI-assisted TCAD design for device-level PPA optimization

Purpose: Give every teammate the same understanding of what is already validated, what will be varied, what must remain fixed, what must be extracted from Sentaurus, and how the TCAD dataset will feed Random Forest/XGBoost, physics interpretation, and NSGA-II optimization.

CURRENT BASELINE SNAPSHOT

Item	Current baseline
Architecture	3-stack Si nanosheet NMOS GAAFET
Baseline geometry	Lg = 10 nm, Wns = 15 nm, Tns = 4 nm
Validated characteristics	ID-VG and ID-VD obtained
Current extracted metrics	SS = 65 mV/dec; DIBL = 10.32 mV/V; Ion = 1.438 mA/um
Immediate goal	Build a controlled geometry DOE and a reliable device-level PPA dataset

Basis: current benchmark comparison report + M.Tech AI project comments.

1. What the project is becoming

The project should now move from a single validated GAAFET simulation to a reproducible design-space experiment. The architecture is frozen first, then selected geometric variables are swept using a planned DOE. Each geometry produces the same electrical outputs, which are converted into device-level power/performance metrics and then used for machine-learning prediction and multi-objective optimization.

Freeze architecture	**DOE geometry**	**Sentaurus TCAD**	**Extract FOMs**	**RF / XGBoost**	**SHAP / physics interpretation**	**NSGA-II optimization**

Key rule: Do not change geometry, physics, bias conditions, extraction definitions, and mesh strategy at the same time. A useful ML dataset requires controlled causes and consistent outputs.

What the comments explicitly recommend

- Start with NMOS and freeze the architecture.
- Use Lg, nanosheet width, and nanosheet thickness as the first geometry variables.
- Generate about 200-250 TCAD geometries using a proper DOE rather than random sampling.
- Extract transistor metrics such as Vth, SS, and Ioff after obtaining reasonable ID-VG and ID-VD characteristics.
- Start ML with Random Forest and XGBoost.
- Hold out specific geometries for testing, then use feature importance / SHAP to connect ML results to device physics.
- Use NSGA-II for multi-objective optimization.

Source: M.Tech AI project comments, pp. 1-2.

2. Phase-1 design variables and frozen architecture

Only three input variables should be varied in the first full experiment. This gives a clean, interpretable dataset and matches the ranges recommended in the project comments.

Variable	Symbol	Range	Proposed 6 levels
Gate length	Lg	10-20 nm	10, 12, 14, 16, 18, 20 nm
Nanosheet width	Wns	15-30 nm	15, 18, 21, 24, 27, 30 nm
Nanosheet thickness	Tns	4-8 nm	4.0, 4.8, 5.6, 6.4, 7.2, 8.0 nm

A 6 x 6 x 6 full-factorial design gives 216 geometries, which lands directly inside the suggested 200-250 geometry range.

Phase-1 DOE size: 216 unique TCAD geometries = 6 Lg values x 6 Wns values x 6 Tns values.

Freeze these initially

- NMOS device type and silicon channel material
- Number of nanosheets = 3
- Sheet spacing = current validated value
- SiO2/HfO2 gate dielectric stack and HfO2 material
- Metal gate and gate work function = current validated value
- Source/drain structure and doping
- Substrate material/doping and channel doping
- BDI structure and thickness
- Corner radius
- Temperature and transport/quantum physics models
- Bias definitions, extraction rules, and mesh refinement philosophy

The comments specifically identify number of sheets, corner radius, channel doping, and work function as variables to consider later, not as mandatory initial variables.

3. Simulation blocks required for every geometry

Each geometry must be run with the same staged Sentaurus workflow so the dataset remains comparable. Preserve the current working SDE/SDevice structure and change only the parameter being swept.

Experiment A - ID-VG at low drain bias Set VDS = 0.05 V. Sweep VGS from 0 to 0.70 V. Use this curve for Vth,lin, SS, Ioff, low-field Ion, and gm.

Experiment B - ID-VG at high drain bias Set VDS = 0.70 V. Sweep VGS from 0 to 0.70 V. Use this curve for Vth,sat, Ion, Ioff, SS, gm, and DIBL calculation.

Experiment C - ID-VD family Sweep VDS from 0 to 0.70 V at VGS = 0, 0.2, 0.4, 0.6, and 0.7 V. Use these curves to observe linear-region behavior, saturation, output conductance, and drive-current dependence.

DIBL definition for the dataset: DIBL = (Vth,lin - Vth,sat) / (VDS,high - VDS,low).

Consistency rules

- Use the same threshold-voltage extraction method for all 216 devices.
- Use the same current normalization convention (for example mA/um) for every run.
- Use identical VGS/VDS sweep endpoints and comparable step sizes.
- Do not retune work function or doping for individual geometries.
- Save raw curves and extracted scalar metrics separately.
- Tag every run with a unique geometry ID so TCAD, extraction, and ML rows cannot be mixed.

4. What we extract: electrical metrics and device-level PPA

Output	Meaning	Why it matters
Vth	Threshold voltage	Switching point and electrostatic design target
SS	Subthreshold swing	Gate control and subthreshold efficiency
DIBL	Drain-induced barrier lowering	Short-channel electrostatic integrity
Ion	ON current	Drive capability / performance
Ioff	OFF current	Leakage and static power
gm,max	Maximum transconductance	Gate control over channel current
Cgg	Total gate capacitance	Needed for intrinsic-delay and dynamic-energy analysis
P_leak	Ioff x VDD	Device-level leakage/static power
tau_int	Cgg x VDD / Ion	Intrinsic delay proxy; lower is better

Power terminology: For this single-device TCAD stage, report leakage/static power clearly. Full dynamic power requires an explicitly defined effective capacitance. The project comments define P(dynamic) = alpha x Ceff x VDD^2 x f and note that Ceff includes gate, diffusion, and interconnect capacitance.

Capacitance terminology must be explicit

Do not use "Cg" ambiguously. Distinguish Cox (oxide capacitance) from Cgg (total small-signal gate capacitance). For the intrinsic-delay metric, Cgg is the more useful quantity if it can be extracted consistently from the chosen Sentaurus setup.

Area terminology

The comments question whether "area" means footprint area. Until a layout-consistent definition is fixed, call it a device-footprint proxy rather than claiming full circuit area. Keep area out of the first optimization objective if its definition is not yet defensible.

Power, delay, gm, area, and capacitance clarification are directly motivated by the project comments on page 1.

5. Practical execution sequence

Do not launch all 216 simulations before proving that each geometric trend behaves physically. The team should build confidence in three small sensitivity sweeps first.

Stage	What changes	What stays fixed	Decision gate
0. Baseline repeat	Nothing	Current validated device	ID-VG/ID-VD and extracted metrics are repeatable
1. Lg sensitivity	Lg = 10-20 nm	Wns = 15 nm, Tns = 4 nm	Trends in Vth, SS, DIBL, Ion are physically reasonable
2. Wns sensitivity	Wns = 15-30 nm	Choose one baseline Lg, Tns = 4 nm	Current scaling and capacitance trends are reasonable
3. Tns sensitivity	Tns = 4-8 nm	Baseline Lg and Wns	Electrostatic/current trends remain stable
4. Full DOE	Lg, Wns, Tns	All architecture/physics settings	216-run dataset passes completeness and quality checks

Recommended first experiment

Gate-length sweep: Run Lg = 10, 12, 14, 16, 18, and 20 nm while keeping Wns = 15 nm and Tns = 4 nm. Extract ID-VG, ID-VD, Vth, SS, DIBL, Ion, Ioff, and gm for all six structures.

Expected physics checks (qualitative)

- As Lg is reduced, short-channel effects should generally become more difficult to control; inspect SS and DIBL carefully.
- As Wns increases, total drive current and gate capacitance can increase; always compare using the same normalization convention.
- As Tns increases, electrostatic control can weaken even if conduction volume rises; this tradeoff is important for optimization.
- Unexpected discontinuities are a reason to inspect geometry, mesh, convergence, extraction, or normalization before accepting the datapoint.

6. Dataset structure and AI workflow

The TCAD dataset should be designed before automation so every simulation creates one consistent machine-learning record plus its associated raw curves.

Input vector X: [Lg, Wns, Tns]

Primary output vector Y: [Vth, SS, DIBL, Ion, Ioff, gm,max, Cgg, P_leak, tau_int]

Dataset block	Example columns	Role	Quality check
Geometry	geometry_id, Lg, Wns, Tns	ML inputs	No duplicates; correct units
Bias/physics metadata	VDD, temperature, model tag	Reproducibility	Same baseline across DOE
Electrical FOMs	Vth, SS, DIBL, Ion, Ioff, gm	Prediction targets	No extraction failures
PPA-derived metrics	Cgg, P_leak, tau_int	Optimization targets	Formula and units fixed
Raw curves	ID-VG low/high VDS, ID-VD family	Audit / curve-level research later	Linked by geometry_id

ML plan recommended by the comments

- Start with Random Forest and XGBoost as strong tabular baselines.
- Split training/testing carefully: hold out one or more specific geometry values rather than relying only on a random row split.
- Report prediction error separately for each target metric.
- Use feature importance and SHAP to check whether the learned relationships make device-physics sense.
- After surrogate validation, use NSGA-II for multi-objective optimization instead of optimizing one metric in isolation.

Source: M.Tech AI project comments, pp. 1-2.

7. Suggested team split and hand-off points

The workflow is easiest to manage when each person owns a reproducible block but uses shared naming, units, and definitions.

Workstream	Main responsibility	Deliverable	Hand-off
TCAD geometry	Parameterized SDE; generate Lg/Wns/Tns structures	Validated geometry + mesh files	To device simulation
Device simulation	Run fixed SDevice physics and bias sweeps	Raw ID-VG / ID-VD outputs	To extraction
Extraction	Automate Vth, SS, DIBL, Ion, Ioff, gm, later Cgg	One clean row per geometry	To dataset owner
Dataset / QA	Merge metrics, metadata, and curve references	Versioned DOE dataset	To ML
ML / physics	RF, XGBoost, hold-out tests, SHAP	Validated surrogate + physics interpretation	To optimization
Optimization / paper	NSGA-II Pareto search and final design selection	Pareto front + selected candidates	Back to TCAD for final verification

Shared naming convention

Use a geometry ID that encodes the three inputs, for example: G_L10_W15_T4p0. Use the same ID in SDE files, SDevice outputs, extraction CSV rows, plots, and ML data. This prevents silent mismatch errors.

Team rule: No teammate changes a frozen parameter or extraction definition without recording the change and rerunning the affected baseline checks.

8. What comes later - not in the first 216-device DOE

The following parameters are valuable for novelty and calibration, but adding them now would expand the design space too quickly and make the first dataset harder to interpret.

Later variable	Why it is interesting	When to add
Number of sheets	Drive current, capacitance, footprint, electrostatics	Phase 2 architecture study
Sheet spacing	Gate fill, electrostatics, parasitics	Phase 2
Corner radius	Field crowding and effective electrostatics	Phase 2
Channel doping	Vth and electrostatics	Phase 2 / calibration
Metal work function	Direct Vth tuning	Phase 2 / calibration
HfO2 thickness / EOT	Gate control and capacitance	Phase 2
BDI material / thickness	Isolation, leakage, electrostatics	Novelty study
Interface traps / contact resistance	Realistic Ioff and Ion calibration	Calibration study
Self-heating / advanced transport	High-field realism	Advanced-physics study

Immediate checklist for the team

1. Freeze the baseline SDE/SDevice files and save a versioned copy.
2. Confirm the exact sheet spacing, dielectric thicknesses, BDI thickness, work function, and doping values that will be frozen.
3. Agree on Vth extraction and current normalization definitions.
4. Run the six-point Lg sensitivity experiment first.
5. Check physical trends and rerun any suspicious datapoint before scaling.
6. Complete Wns and Tns sensitivity sweeps.
7. Launch the 216-point DOE only after the above passes.
8. Build extraction automation and dataset QA in parallel with TCAD runs.
9. Start RF/XGBoost only after the dataset schema is frozen.
10. Use SHAP/feature importance and NSGA-II after surrogate validation.

Next concrete task: Experiment 1 = gate-length sweep. Change only Lg. Keep Wns = 15 nm, Tns = 4 nm, three sheets, and all validated physics/mesh/bias definitions fixed.

Source basis for this team guide

This guide is intentionally grounded in the two project documents already reviewed in the team discussion. It does not introduce a new literature benchmark; its purpose is to convert the supervisor/project comments into an executable experiment plan.

Document	What was used
Mtech AI project comments(2).pdf	Feasibility/novelty comments; initial variables; recommended ranges; static/dynamic power definitions; delay/gm/area/Cg questions; RF/XGBoost; 200-250 DOE geometries; fixed architecture; train/test hold-out; NSGA-II; SHAP/feature importance.
GAAFET_3NS_Benchmark_Comparison_Report(1).pdf	Current device baseline and previously extracted TCAD performance metrics used only to summarize project status on the cover page.

Prepared as an internal execution guide for the GAAFET TCAD + AI team.